Class-01-Intro-GenAI-LLMs

🔹 Check UV installed

```
uv --version
```

🔹 Install UV (if not installed)

```
pip install uv
```

🔹 Initialize Project

```
uv init
```

🔹 Check Installed Packages

```
uv pip list
```

🔹 Check Available Python Versions

```
uv python list
```

🔹 Create Virtual Environment

Specific version:

```
uv venv env --python cpython-3.12.18-windows-x86_64-none
```

Generic:

```
uv venv <your-env-name> --python <your-python-version>
```

🔹 Install Packages

Single package:

```
uv add <package_name>
uv pip install <package_name>
```

From requirements file:

```
uv add -r requirements.txt

uv pip install -r requirements.txt
```

LLM / GenAI Development Tools & Frameworks

LLM Foundations

- Huggingface
- PyTorch

Finetuning

- Unsloth
- LLaMA Factory
- Huggingface

Vector Databases & Search (RAG Backbone)

- FAISS
- Chroma
- Qdrant
- Supabase

- pgvector
- Milvus
- Pinecone
- AWS OpenSearch
- Azure AI Search

Databases & Storage

- Neo4j
- MongoDB
- PostgreSQL
- Redis
- SQL Alchemy

LLM Application Building Framework

- LangChain
- Llama Index
- Haystack
- Lang Smith

Guardrails, Validation & Safety

- Guardrails.ai
- Nvidia Nemo
- OpenAI Guardrail

Local LLMs & Runtime

- Ollama

Experiment Tracking

- MLflow

Data Modelling

- Pydantic

MCP

- FastMCP

Document AI & Parsing

- Llama Parser
- Docling
- Google Document AI
- Amazon Textract
- python-docx
- PyMuPDF
- pdfplumber
- Parsio
- Unstructured.io
- Azure Document Intelligence

Agentic AI Frameworks

- Lang Graph
- Deepagent
- AutoGen
- n8n

Cloud GenAI Platforms

- Google ADK
- Agent Core
- AWS Bedrock
- Azure Cloud Foundry
- GCP Vertex AI

Evaluation & Observability

- RAGAS

- TruLens
- Llama Index Eval
- DeepEval
- OpenAI Evals

Prompt Engineering & Prompt Management

- LangChain Prompt Hub
- **PromptLayer**
- OpenAI Cookbook / Prompt Examples
- PromptHub (Community-driven repos)
- DSPy (Prompt-as-Code approach)
- Guidance (Microsoft)

Deployment, Infrastructure & CI/CD

- ECS + Fargate + EC2
- AWS SageMaker
- GitHub Actions
- Azure Virtual Machines (VMs)
- Azure Kubernetes Service (AKS)
- Azure Container Apps / ACI

Testing

- pytest

Pipeline Orchestration

- Apache Airflow